

1. List 5 common issues users face with USB keyboards or mice on laptops, and write one possible solution for each.

1. The Device is Completely Unresponsive

- Sometimes you plug in your mouse or keyboard, and absolutely nothing happens—no lights, no cursor movement, no typing.
- **Solution: Swap the USB Port.** It sounds simple, but USB ports on laptops can fail or temporarily glitch out. Unplug the device and plug it into a different port on the opposite side of the laptop. If it works, the original port might be dirty or damaged. If it still doesn't work, try plugging the device into a completely different computer to see if the hardware itself is dead.

2. Random Disconnects and Reconnects

- Your device works fine, but suddenly it disconnects (often accompanied by the classic Windows USB disconnect chime) and then reconnects a few seconds later.
- **Solution: Disable USB Selective Suspend.** Laptops try to save battery by cutting power to USB ports they think aren't being used. Sometimes, this feature is too aggressive.
- *On Windows:* Go to **Control Panel > Power Options > Change plan settings > Change advanced power settings**. Expand the **USB settings** and disable **USB selective suspend setting**.

3. Laggy Movement or Delayed Typing

- You move your mouse, but the cursor trails behind, or you type a sentence and the letters appear a second later. This is incredibly common with wireless USB devices (the ones that use a little receiver dongle).
- **Solution: Eliminate Interference and Check Power.** First, swap out the batteries in your mouse or keyboard for fresh ones. If the lag persists, move the USB dongle to a different port. USB 3.0 ports (often blue inside) can emit wireless interference that messes with 2.4GHz wireless dongles. Moving the receiver to a USB 2.0 port or using a short USB extension cable to distance it from the laptop often cures the lag.

4. "USB Device Not Recognized" Error

- You plug the device in, and your operating system immediately throws up a pop-up saying the device malfunctioned or isn't recognized.
- **Solution: Reinstall the Drivers.** Software communication between the laptop and the device has likely corrupted.
- *On Windows:* Right-click the Start button and open **Device Manager**. Find the unrecognized device (it will usually have a yellow warning triangle next to it), right-click

it, and select **Uninstall device**. Unplug the USB, restart your laptop, and plug it back in. The computer will automatically grab a fresh installation of the driver.

5. Mouse Cursor is Erratic or Keys Type the Wrong Things

- **Solution: Clean the Hardware and Check OS Settings.** * *For the mouse:* Flip it over and check the optical sensor. A single piece of dust or a stray hair trapped in the sensor hole will cause the cursor to jump wildly. Blow it out or gently clean it with a cotton swab.
- *For the keyboard:* If it's typing the wrong characters, your laptop's keyboard language layout has likely been accidentally changed (often triggered by pressing Alt + Shift or Windows key + Spacebar). Check your system settings to ensure your preferred language and layout (like US QWERTY) are selected.

2. Simulate a troubleshooting flow: Your Bluetooth headphones are not connecting to your phone. Write the exact step-by-step actions you would take to diagnose and fix the issue, mentioning any settings or indicators you would check.

Here is the exact troubleshooting flow I would follow if my Bluetooth headphones were not connecting to my phone:

1. Verify the headphones are powered on

1. Press the power button on the headphones.
2. Check for any power indicator light.
3. Confirm the battery is not depleted:
4. If the battery indicator shows low power, connect the headphones to a charger for at least 10–15 minutes.
5. Check whether a charging LED appears.

2. Confirm Bluetooth is enabled on the phone

1. Open **Settings**.
2. Tap **Bluetooth**.
3. Verify that the Bluetooth toggle is switched **ON**.
4. Wait a few seconds to see whether the headphones appear in the device list.

3. Check whether the headphones are already connected elsewhere

1. Think about other devices previously paired with the headphones (laptop, tablet, another phone, TV, etc.).
2. Turn Bluetooth off on those devices temporarily.
3. Retry the connection from the phone.

4. Put the headphones into pairing mode

1. Turn the headphones off.
2. Press and hold the pairing/power button according to the manufacturer's instructions (often 5–10 seconds).
3. Watch for pairing indicators:
 - Flashing blue light
 - Alternating red/blue light
 - Voice prompt such as "Pairing" or "Ready to connect"
4. Verify the headphones now appear under **Available Devices** on the phone.

5. Attempt a fresh connection

1. On the phone's Bluetooth screen, tap the headphone name.
2. Wait for the status to change to:
 - **Connected**
 - **Connected for calls and audio**
3. Test by playing audio.

6. Forget and re-pair the device

If the headphones appear but won't connect:

1. Open **Settings** → **Bluetooth**.
2. Find the headphone entry.
3. Tap the information icon (ⓘ, gear icon, or similar).
4. Select **Forget Device**, **Unpair**, or **Remove Device**.
5. Confirm removal.
6. Put the headphones back into pairing mode.
7. Scan for devices again and reconnect.

7. Restart both devices

1. Turn the headphones off.
2. Restart the phone completely.
3. Turn the headphones back on.
4. Place them in pairing mode.
5. Try connecting again.

8. Check Bluetooth permissions and settings

1. Verify Bluetooth is not restricted by:
 - Airplane Mode
 - Battery Saver mode
 - Device management policies
2. Turn Airplane Mode on for 10 seconds and then off again.
3. Return to Bluetooth settings and retry.

9. Test Bluetooth functionality

1. Try connecting a different Bluetooth device to the phone.
2. Try connecting the headphones to a different phone.
3. Interpret the results:
 - If other devices connect to the phone, the issue is likely with the headphones.
 - If the headphones connect to another phone, the issue is likely with the original phone.

10. Reset the headphones

If connection problems continue:

1. Locate the headphone reset procedure in the manual.
2. Perform a factory reset (often holding multiple buttons for several seconds).
3. Watch for a reset confirmation indicator:
 - LED flashing pattern
 - Voice prompt
4. Pair the headphones with the phone again.

11. Update software

1. Check for phone operating system updates:
 - **Settings** → **Software Update/System Update**
2. Install any available updates.
3. If the headphone manufacturer provides an app, check for firmware updates for the headphones.
4. Restart and test again.

12. Final verification

1. Confirm the headphones show **Connected** in Bluetooth settings.
2. Play music or a video.
3. Verify audio comes through the headphones.
4. Test volume controls and microphone functionality.

If all of these steps fail, I would suspect a hardware fault in either the headphones' Bluetooth module or the phone's Bluetooth radio and would test with additional devices or contact the manufacturer's support team.

3. **A user reports that their external webcam is not detected during a Zoom call on Windows 10. Write a checklist of 6 things to check or try to resolve the problem.**

External Webcam Not Detected in Zoom (Windows 10) – Troubleshooting Checklist

1. Check the physical connection

- Ensure the webcam is securely plugged into a USB port.
- Try a different USB port on the computer.
- If using a USB hub, connect the webcam directly to the PC.

2. Verify Windows detects the webcam

- Open **Device Manager**.
- Expand **Cameras** or **Imaging Devices**.
- Confirm the webcam appears and does not show a warning icon.
- If it is missing, unplug and reconnect the device.

3. Check camera privacy settings

- Go to **Settings** → **Privacy** → **Camera**.

- Make sure **Camera access for this device** is enabled.
- Ensure **Allow desktop apps to access your camera** is turned on.

4. Select the correct camera in Zoom

- Open Zoom and go to **Settings** → **Video**.
- Check the **Camera** dropdown menu.
- Select the external webcam instead of the built-in laptop camera.
- Verify that a video preview appears.

5. Close other applications using the camera

- Exit applications such as Teams, Skype, OBS, or browser tabs that may be using the webcam.
- Reopen Zoom and test the camera again.

6. Update or reinstall webcam drivers

- In **Device Manager**, right-click the webcam and select **Update driver**.
- If issues persist, uninstall the device and restart the computer.
- Windows will typically reinstall the driver automatically after reboot.

4. You are helping a friend whose printer is showing 'offline' status on their PC. Describe how you would use the Windows Troubleshooter tool to try to fix the issue, and explain what information the tool provides during the process.

If a printer is showing **Offline** in Windows, I would first try the built-in **Windows Troubleshooter**, which can automatically detect and fix common printer problems.

Steps to run the Printer Troubleshooter

1. Click **Start** and open **Settings**.
2. Go to **System** → **Troubleshoot** → **Other troubleshooters** (on Windows 11).
 - On Windows 10, go to **Settings** → **Update & Security** → **Troubleshoot** → **Additional troubleshooters**.
3. Find **Printer** and click **Run**.
4. Select the printer that is showing as offline if prompted.

5. Allow Windows to scan for issues and follow any recommendations it provides.

What the Troubleshooter does

During the process, the tool checks for common problems such as:

- Whether the printer is set as the default printer.
- Whether the printer is paused or marked as “Use Printer Offline.”
- Problems with the print spooler service.
- Missing or outdated printer drivers.
- Connectivity issues (USB, Wi-Fi, or network-related problems).

Information provided by the tool

As it runs, the Troubleshooter displays:

- **The diagnostic steps being performed**, such as checking the printer queue or restarting services.
- **Problems found**, for example, “Print spooler service is not running” or “Printer is set to offline.”
- **Actions taken automatically**, such as restarting the print spooler or changing printer settings.
- **Recommended manual steps** if a problem cannot be fixed automatically.
- **A summary report** showing whether issues were fixed, not fixed, or if further action is needed.

After the troubleshooter finishes, I would try printing a test page to see if the printer is back online. If the problem persists, the report from the Troubleshooter helps identify the next steps, such as reinstalling the driver or checking the printer’s network connection.